IMPROVED PREDICTION OF GAS HYDRATE SATURATION USING MACHINE LEARNING AND OPTIMAL SET OF WELL-LOGS

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Abstract

Resistivity and acoustic logs are widely used to estimate gas hydrate saturation in various sedimentary systems using one of the two popular methods (1. Acoustic velocity, and 2. Electrical resistivity), but the limitations of these two methods are often overlooked, which include i) well-specific calibration of empirical exponents in the electrical resistivity method, ii) assumption of known pore morphology for gas hydrates in the acoustic velocity method, iii) presence of unknown mineralogy and bulk modulus terms in the acoustic velocity method. NMR-density porosity derived gas hydrate saturation based on the analysis of the transverse magnetization relaxation time ($T2$) is considered the most precise method, but acquisition of NMR-based logs is limited at relatively recent drilled sites; additionally, its use in conventional oil and gas reservoirs is not that common due to higher cost and operational deployment limitations associated with acquiring NMR well-logs. This study proposes a new method that predicts gas hydrate saturation (S_h) for any well using porosity, bulk density, and compressional wave (P-wave) velocity well-logs with Neural Network (or Stochastic Gradient Descent Regression) without any well-specific calibration and/or other aforementioned shortcomings of the existing methods. The method is developed by examining the underlying dependency between S_h and different combinations of well-logs, chosen from 6 routine logs, with 12 different machine learning (ML) algorithms. The accuracy of the proposed method in predicting S_h is ~84%, which is better than the accuracy of seismic and electrical resistivity methods ($\leq$ 75%) reported by three different studies. The robustness of the method in the specific case of permafrost associated gas hydrates is demonstrated with well log data from two wells drilled on the Alaska North Slope.

Keywords: machine learning; gas hydrates; well-logs; neural network; supervised learning

1. Introduction

Well-logs are rock and fluid measurements obtained through tools that are run down through a deep wellbore during or after drilling to quantify reservoir properties and evaluate its potential. Some of the well-logs are routinely acquired for all the wells, whereas some advanced well-logs like nuclear magnetic resonance (NMR) are not part of the commonly acquired well-logs due to higher cost associated with their acquisition as well as operational challenge in deploying the NMR tool in the past [1]; however, some of the challenges have since been overcome with the introduction of newer version of NMR tools (*e.g.* CMR and ProVision by Schlumberger). The difficulties in acquiring NMR log formed the motivation for some studies to use ML to predict formation properties [2–4] that are otherwise available only through NMR logs. Although the experience of acquiring NMR logs in gas hydrate reservoirs for research purposes does not exhibit similar difficulties, these problems cannot be ruled out completely as those difficulties are the primary motivation behind a recent study [4]. Other uses of ML in the oil and gas industry have been automation of subjective and time-consuming workflows to improve efficiency, consistency of workflows, and objectivity in applications such as well-log processing/interpretation [5,6], selecting enhanced oil recovery method [7–10], and well-based operations [11]. ML has also been used to address the inadequacy of classical physics-driven and empirical models when applied to shales, such as predicting time-series forecast of oil production rates [12–15], and predicting bottomhole pressure [16,17].

All the above discussed reasons to adopt ML in exploration and production of hydrocarbons have equal relevance in gas hydrate reservoirs, which is a potentially significant gas resource found worldwide in permafrost regions and marine environments, with a potential to supply energy at global-scale in future. One key parameter required to evaluate the resource and performance of gas hydrate reservoirs is gas hydrate saturation (S_h), which can be estimated using empirical Archie's law, acoustic velocity method, and NMR-density porosity method by processing resistivity log, sonic well-logs, NMR and density well-logs, respectively, along with other data (porosity, mineralogy, bulk modulus).

In this study, ML is used to predict S_h using an efficient method, which is not constrained by the assumptions and unknown parameters of the physics-driven methods (e.g. acoustic velocity method) or empirical (e.g. Archie's law) models; additionally, the proposed method to predict S_h does not require NMR well-log. Twelve different ML algorithms were employed to examine the underlying dependency between S_h as revealed from NMR data (herein assumed to be the "true" S_h), and different combinations of routine well-logs from gamma ray (GR), rock bulk density (ρ_b), neutron porosity (ϕ_{neut}), density porosity (ϕ_{den}), resistivity (R_t), and sonic compressional velocity (v_p). The robustness of the proposed method is demonstrated with respect to permafrost gas hydrate with well log data from two wells drilled on the Alaska North Slope as the data for these wells are available in the public domain.

The method's approach, machine learning algorithms, and data used in developing the method are discussed next. Results section presents two case studies that illustrate the application of the method, and the results from the two cases are then used to optimize the method. The results are followed by a discussions section, following which the proposed study is summarized and concluded.

2. Method

2.1. Existing Methods to Estimate Gas Hydrate Saturation

Gas hydrate saturation can be estimated using several empirical and rock and pore fluid-physics based relationships [18–23] that are functions of physical properties, and may also depend on the morphology of gas hydrates (e.g. pore filling, grain coating, etc.) within the host sediments.

2.1.1. Acoustic Velocity Method

In this method, acoustic velocities (compressional and shear wave velocities) are used to estimate S_h using relationships that are either empirical [18] or based on rock-physics effective medium theory [19,24]. Commonly used empirical relationships are Wyllie's time-average relation, Wood's relation, and their variants [18], whereas commonly used rock-physics based relationships are Dvorkin's relation [25], and their variants [19,21].

Acoustic velocity models have two major limitations, which are i) presence of unknown terms (mineralogy and bulk modulus) that must be acquired through other means, and ii) assumption of uncertain pore morphology (e.g. pore filling, grain coating, load bearing, etc.) for gas hydrates. Theoretical models that describe effective properties of gas hydrates assume one or more uncertain pore morphology characteristics [26–30] of hydrates, which is a difficult information to obtain even in laboratory-scale samples [31], whereas the pore morphology of hydrates cannot be measured directly in the field. The pore morphology of hydrate is generally a combination of various morphologies that transitions from one morphology to another with varied grain sizes or its saturation, but also perhaps due to charging mechanism and history between different accumulations of gas hydrates. The presence of any free gas (metastable) in the gas hydrate zone also limits the usefulness of acoustic methods. Acoustic method may not give exact numbers for hydrate saturation, but it can place bounds on gas hydrate saturation. Nevertheless, the sonic logs are very useful to understand the location of hydrate, and the type of hydrate present based on the v_p and v_s values and also the nature of the sonic waveforms; typically showing high values of attenuation.

2.1.2. Electrical Resistivity Method

Gas hydrates act as electrical insulator, so the presence of gas hydrate-bearing sediments increases the resistivity of rock. This information is used with Archie's law [32] to estimate gas hydrate saturation [33] as follows:

$$S_h = 1 - S_w = 1 - \left(\frac{R_0}{R_t}\right)^{\frac{1}{n}} \tag{1}$$

Where, R_0, R_t are resistivity of reservoir rock saturated with water, and log-measured resistivity of reservoir rock saturated with all in-situ fluids, respectively.

Generally, R_0 can be calculated using R_w, which is resistivity of connate water (without rock matrix), as $R_0 = F.R_w = \left(\frac{a}{\phi^m}\right).R_w$ that gives:

$$\Rightarrow S_h = 1 - \left(\frac{a}{\phi^m}\right)^{\frac{1}{n}}.\left(\frac{R_w}{R_t}\right)^{\frac{1}{n}} \quad (2)$$

Here, a, m, n are empirical parameters, where m is a function of rock cementation and independent of hydrate saturation, while n is a function of hydrate morphology. Generally, the value of the factor a is considered to be 1.

The main requirement [32,34] to use Archie's law is that the reservoir rock does not contain very low-salinity formation water such that surface conductivity is significant. Besides these requirements that are suggested to be met [32,34] in order to use Archie's law, a major limitation of using this method to calculate S_h is calibration of empirical parameters (m, n) that may be different for each well, which is possible only using S_h that is already known. Some studies (e.g. Cook and Waite 2018) suggest estimating S_h using a potential universality in one of these three parameters (n), lying between 2.0 to 3.0, but such approach is limited for practical applications because it is only applicable to coarse-grained reservoir with saline pore water. Specifically, even in coarse-grained rocks the uncertainty in n (lying between 2.0 to 3.0) can lead to errors as large as 50% for moderately high value of S_h (~0.5), whereas the error in S_h can jump over 100% for other inter-bedded layers with clay/silt and other fine-grained minerals. Besides, the potential universality approach does not consider uncertainty in the other empirical parameter m that may further introduce errors over and beyond the existing errors due to uncertainty in n. Modified form of Archie's method developed to work for shaly sand hydrocarbon reservoirs work just as well for gas hydrate. Resistivity based methods, in general are more reliable and accurate way to detect and quantify gas hydrates when compared with the other methods. Resistivity tools function can be relatively insensitive to wellbore condition, and some measures of anisotropy such as layering, and even fracture sets, can be detected and accounted for.

2.1.3. NMR-Density Porosity Method

NMR logging tool measures water-filled porosity, whereas density porosity measures the pore space by water and/or gas hydrate [22,36,37] such that the difference between the density porosity and NMR porosity is the portion of pore space filled with just gas hydrate, which can be used to estimate S_h as follows [23,36,38–40]:

$$S_h = 1 - \frac{\phi_{nmr}}{\phi_{den}} \quad (3)$$

Although well-log tools in many gas hydrate-bearing sediments operate in low density environments where the calibration of density and neutron tools is suspect, or out of range, the NMR method that depends on the tool used is in general more reliable. The main technical limitation of NMR is high temperature, which is not encountered in gas hydrate wells, but the major barrier of this method beyond the scientific consideration is that the acquisition of advanced well-logs like NMR is limited at relatively recent drilled sites, which is exhibited by its limited acquisition in commercial oil and gas fields [2–4].

The summary of above discussed methods and their comparison with the proposed ML-based method is presented in Table I.

Table I: Comparison of three different methods to estimate gas hydrate saturation with our ML-based method.

Method Name	***Description***	***Pros***	***Cons***
Acoustic Velocity	Acoustic log velocities are used to estimate S_h using one of several empirical or rock-physics based relationships.	Effective medium theory-based mechanistic model.	i) Requires hydrate morphology, which is typically not known ii) mineralogy and bulk modulus terms are unknown or open to interpretation from various data sources.
Electrical Resistivity	Archie's law to estimate S_h.	Simple equation.	i) assumption of no highly tortuous pore system, ii) no thinly laminated reservoirs, iii) well-specific calibration for three unknown parameters.
NMR-Density Porosity	Density porosity and NMR porosity are used to estimate S_h.	i) simple equation, ii) no well-specific calibration.	Limited acquisition compared to commonly acquired well-logs, possibly due to cost and operational deployment difficulties.
This study	Predicts S_h using just three simple well-logs (ϕ_{neut}, ρ_b, v_p) with ML.	i) No well-specific calibration, ii) can be used to fill the gaps where NMR data is not available, iii) no assumed parameters.	i) Cannot provide pre-drilling information.

2.2. Proposed Method to Estimate Gas Hydrate Saturation

The proposed method assumes that i) NMR-density porosity derived S_h is the ground truth, ii) S_h has an underlying dependency or high correlation with some well-known parameters available from routine well-logs, which are GR, ϕ_{neut}, ϕ_{den}, ρ_b, R_t, v_p. Using this assumption as the basis of the proposed method, ML is used to examine the relationship between S_h and 24 different well-log combinations (WLC) chosen from the 6 logs mentioned above. Prior to using the well-logs for analysis in the general workflow, they were pre-processed for depth-matching (due to different resolutions of various well-logs), and identification of badhole where there might be a washout or swell in the wellbore. To identify a badhole, following condition was used at each depth: $(caliper_{min} < CALD[i] < caliper_{max})$ and $(CORR[i] < DRHO_{max})$, where $CALD[i]$ and $CORR[i]$ indicate caliper and density correction log values at depth index i. The value of $caliper_{min}$ and $caliper_{max}$ are based on the 10th percentile value of the caliper log ($CALD_{P10}$) and the wellbore radius, whereas $DRHO_{max}$ is the maximum value of the density correction well-log.

The general workflow and the investigation to obtain optimal WLC and the optimal ML model are summarized in Figure 1. A combination of few well-logs out of the 6 logs is selected and each well-log is divided into three sets along the depth as a training set, a validation set (a subset of the training set), and a test set, respectively, including the corresponding sections for ground truth S_h (NMR-density porosity derived S_h) of that well. A standard ML process involves three datasets, which are training set, validation set, and testing set. The ML model is optimized for its hyperparameters, which are parameters whose values are not derived via training and have to be set prior to training. In this study, the hyperparameters are optimized using grid search, which is an exhaustive search through a pre-set space of hyperparameters. The ML model is then used to predict S_h for another well, known as a 'blind well', which is independent of the well that trained the ML model to ensure that the proposed method is able to predict S_h for any new well without calibration and training. The validation set (also called "dev" set) is primarily used to tune the hyperparameters of a ML algorithm through a cross-validation splitting of the training dataset, and the test set is used as a test of prediction capability of the algorithm. Although considering both validation and test sets helps to evaluate the strength of a ML algorithm, it is not a customary practice to consider both validation and test sets when the availability of data is an issue [4]. However, the key test of prediction capability in this study is served by a 'blind well', which is fixed in length and establishes the performance of the proposed method in a truly unbiased manner. Therefore, unlike the traditional purpose of the validation and test dataset, in this study the validation and test datasets from the same well are used to partially verify the optimal solution for WLC, because unlike a 'blind well' that serves as a complete and independent test, the validation and test datasets that are part of the training well would not depict the holistic performance of the optimal WLC. The proposed method to predict S_h does not require retraining when predicting S_h for an unrelated ('blind') well; in other words, the method is a ready-to-use trained ML algorithm that can predict S_h for any unrelated target well without retraining.

The general workflow, summarized in Figure 1, is used within a framework that also includes finding an optimal WLC (out of 24 combinations shown in section 2.4) and an optimal ML algorithm (out of 12 algorithms), which result in best prediction of S_h for the blind well.

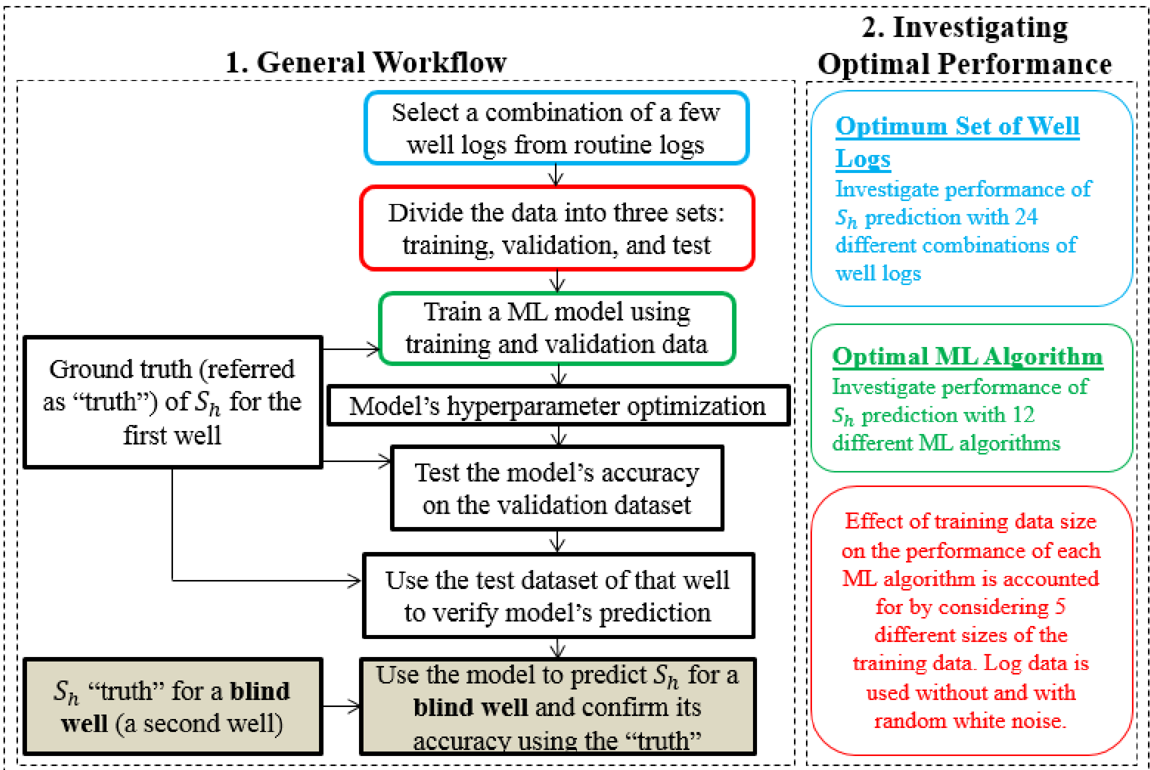

Figure 1: Summary of the approach used in developing the proposed method.

2.3. Supervised Learning Algorithms

Different ML algorithms work better for some problem types than others, and despite the theoretical understanding of the algorithms in terms of their limitations and advantages, their relative performance on a particular dataset cannot be ranked without examining them practically. In this study, 12 different supervised ML algorithms belonging to 5 different classes, summarized in Table II, are used to predict S_h. The first class of algorithm is Ridge Regression [41] and its variants, including Kernel Ridge Regression (KRR) [42], Support Vector Regression (SVR) [43], and Stochastic Gradient Descent Regression (SGDR) [44]. The second class of algorithm is Decision Tree [45] and its variants, including Random Forest [46], Adaptive Boosting (AdaBoost) [47], and Gradient Boosting [48]. The third and fifth classes of algorithms are with one algorithm each, including k-Nearest Neighbor [49] and Multilayer Perceptron that is also known as Neural Network [50], respectively. The fourth class of algorithm is reduced order model variants, including multivariate adaptive regression splines (MARS) [51] and generalized additive model (e.g. polynomial regression) [52]. These algorithms were implemented using Python and its machine learning packages. A desirable characteristic of a ML algorithm is its ability to generalize, which means that the algorithm should have the ability to avoid over-fitting. Although some ML algorithms have the in-built theoretical capability to avoid over-fitting through specific hyperparameters, some algorithms that do not have this inherent advantage can be checked for over-fitting through cross-validation as well as a two-step process that involves checking the model's score for accuracy during training and validation, such that the score on the validation dataset should be comparable to the score on the training dataset with a reasonable margin of error; this check for over-fitting can be extended further to an unseen data, which in this study is exhibited through a 'blind well'.

Table II: Summary of supervised ML algorithms used in this study to predict S_h.

Class	***Algorithm***	***Description***	***Over-fitting Handling Capability***
Ridge Regression and its Variants	Ridge Regression	Linear least square with L2-norm regularization.	Overfitting is handled through a regularization parameter. Regularization (e.g. L1, L2 weight penalties) significantly reduces the variance of the model, while also not increasing its bias substantially.
	KRR	Combines ridge regression with a kernel to induce learning.	
	SVR	It differs from KRR in terms of the loss function such that it only uses a subset of the training data, and ignores any training data that lie within a margin of the model prediction.	
	SGDR	The gradient of the loss is estimated for each sample at a time and the model is updated along the way, while also decreasing the strength of the learning rate that updates the model.	
Decision Tree and its Variants	Decision Tree	Non-parametric supervised learning where value of a target variable is predicted by learning simple decision rules, inferred from the data features (e.g. sine curve) to map input with an output using a single tree.	Although the fundamental form of decision tree does not have any in-built theoretical capability that prevents over-fitting, the concept of ensembled tree (e.g. Random Forest, AdaBoost, Gradient Boosting)
	Random Forest	It fits a number of decision trees on various sub-samples of the dataset and uses averaging to improve prediction and control over-fitting.	
	AdaBoost	Used with decision tree to do recursive regression on the same dataset by adjusting the weights of instances per the error of current prediction.	

	Gradient Boosting	Builds an additive model in forward stages by fitting a regression tree on the negative gradient of the loss function.	helps prevent over-fitting.
k-Nearest Neighbor		Regression is based on k-closest training data that do not create a general internal model, but store instances of the training data to predict the target by local interpolation of the associated nearest neighbors.	Over-ftting in the kNN algorithm is controlled using the value of parameter k that is tied to the error rate of the model.
Reduced Order Models	Polynomial Regression	Transforms the input data into a polynomial combination of the input features with a degree that is less than or equal to the specified degree. Then linear regression is performed on the polynomial developed from the input features.	Over-fitting can be controlled using the degree of polynomial, such that the higher-order polynomial tend to over-fit.
	MARS	Weighted sum of basis functions, which can be a constant, a hinge function, and/or a product of more than one hinge functions. It uses backward pass to control an over-fitted model by pruning the model developed in the forward pass.	Over-fitting is controlled via backward pass by iteratively pruning the basis functions that contribute least to the fit.
Neural Network	Multilayer Perceptron	It learns a function through 1 or more hidden layers between the output and input layers. Each layer contains multiple hidden neurons that transform the values from previous layer with a mapping function that is a weighted linear summation followed by an activation function (e.g. arc tan).	Overfitting is handled through a regularization technique called 'dropout' that iteratively drops fraction of weights.

2.4. Well-log Combinations

The proposed method assumes that S_h has an underlying dependency with some parameters available as routine well-logs (GR, ϕ_{neut}, ϕ_{den}, ρ_b, R_t, v_p), but it is not known *a priori* what combination of these logs is strongly linked to S_h. Although conventional physics-based methods to predict S_h suggest that R_t and v_p respond directly to the presence of gas hydrate, this study does not invoke any physics in selecting the well-logs. Additionally, other ML-based studies [53] may start with one pre-defined combination of well-logs that may not give the best outcome in terms of prediction, as this study shows. Using the assumption of the underlying dependency as the basis of the proposed method, ML is used to examine the underlying dependency between S_h and arbitrarily chosen 24 different well-log combinations (WLC) out of all the possible combinations, shown in Table III, chosen from the 6 routine logs mentioned above. Considering all the possible combinations from 6 well-logs would be a large number to manage, while at the same time many of those combinations perform equally bad such that considering those combinations may not add value to the method. Therefore, the 24 different WLC chosen for this study have been partially hand-crafted in a way that half of the 24 WLC contain at least 3 well-logs with one or more primary variables (*i.e.* porosity, resistivity, and compressional velocity; the variables used to predict S_h through NMR-density, Archie's law, and acoustic velocity methods, respectively) whereas the other half of 24 WLC combinations contain a single or two well-logs to investigate whether S_h can be reasonably predicted with only one or two well-logs. The order of WLCs shown in Table III does not depict any preference or likelihood of that particular WLC in predicting S_h, and the number assigned to each combination of well-logs is simply a programmatic outcome.

ML solves any problem by recognizing patterns that exist between the input and the output data; however, a pattern may not necessarily exist between each feature (*e.g.* each well-log) of the input data and the output data (*e.g.* S_h), in which case the ML algorithm may not be able to accurately predict the output. Therefore, to optimally exploit the performance of ML in predicting the output (S_h), an optimal WLC that has an optimal underlying pattern with S_h is investigated.

Table III: Twenty four different well-log combinations from the 6 routine logs.

WLC #	***Well-Logs***
1	$\phi_{den},\ R_t$
2	$GR,\ \phi_{neut},\ \rho_b$
3	R_t
4	$R_t,\ v_p$
5	$\phi_{neut},\ \phi_{den},\ \rho_b,\ R_t,\ v_p$
6	$\phi_{den},\ \rho_b,\ v_p$
7	ϕ_{neut}
8	$\phi_{neut},\ \rho_b,\ v_p$
9	ρ_b
10	$GR,\ \phi_{neut}$
11	$GR,\ \phi_{den},\ \rho_b,\ R_t,\ v_p$
12	$\phi_{neut},\ \rho_b,\ R_t$
13	$GR,\ R_t$
14	$GR,\ \rho_b$

15	$GR,\ v_p$
16	$\phi_{den},\ \rho_b,\ R_t,\ v_p$
17	$GR,\ \phi_{neut},\ \rho_b,\ R_t,\ v_p$
18	$\phi_{neut},\ \rho_b,\ R_t,\ v_p$
19	$\rho_b,\ R_t,\ v_p$
20	$\phi_{den},\ \rho_b,\ R_t$
21	ϕ_{den}
22	v_p
23	GR
24	$GR,\ \phi_{neut},\ \phi_{den},\ \rho_b,\ R_t,\ v_p$

2.5. Known Field Data

For the purpose of this study, six routine logs ($GR,\ \rho_b,\ \phi_{neut},\ \phi_{den},\ R_t,\ v_p$) are available from two different wells (Mt. Elbert and Ignik Sikumi test wells, respectively) described next, whereas the ground truth for S_h that will be used to verify the ML-predicted S_h is measured using NMR porosity (ϕ_{nmr}) and density porosity (ϕ_{den}) presented earlier, which does not depend either on the reservoir model or on any other parameter that requires calibration [23], except for assuming that the gas hydrate, not massive, is in the pore space and the grain density is constant. Therefore, the estimation of S_h using NMR-density porosity method is most precise and accurate of other methods; hence, this method is used to obtain the "ground truth" of S_h. Considering S_h measured using NMR porosity (ϕ_{nmr}) and density porosity (ϕ_{den}) as the ground truth is an accepted practice, which has been used by others [22,23,36] to assess the accuracy of existing methods (Archie's equation and acoustic velocity method) in predicting S_h. The use of ϕ_{nmr} in this study is only limited to verify accuracy of the proposed method, and ϕ_{nmr} is not required to predict S_h by the proposed method.

2.5.1. Mt. Elbert Stratigraphic Test Well ("Well-MTE")

A joint collaboration between BP Exploration (Alaska), the U.S. Department of Energy, and the U.S. Geological Survey conducted a well test in 2007 that was called 'Mt. Elbert Stratigraphic Test Well' [54,55]. The data acquired for this well, located at Alaska North Slope, include suite of well-logs, coring, seismic, and pressure testing. The six routine logs ($GR,\ \rho_b,\ \phi_{neut},\ \phi_{den},\ R_t,\ v_p$), and ground truth for S_h estimated using NMR log (ϕ_{nmr}) are shown in Figure 2. The depths of investigation range from 2006.5 ft to 2206 ft with 822 number of data points.

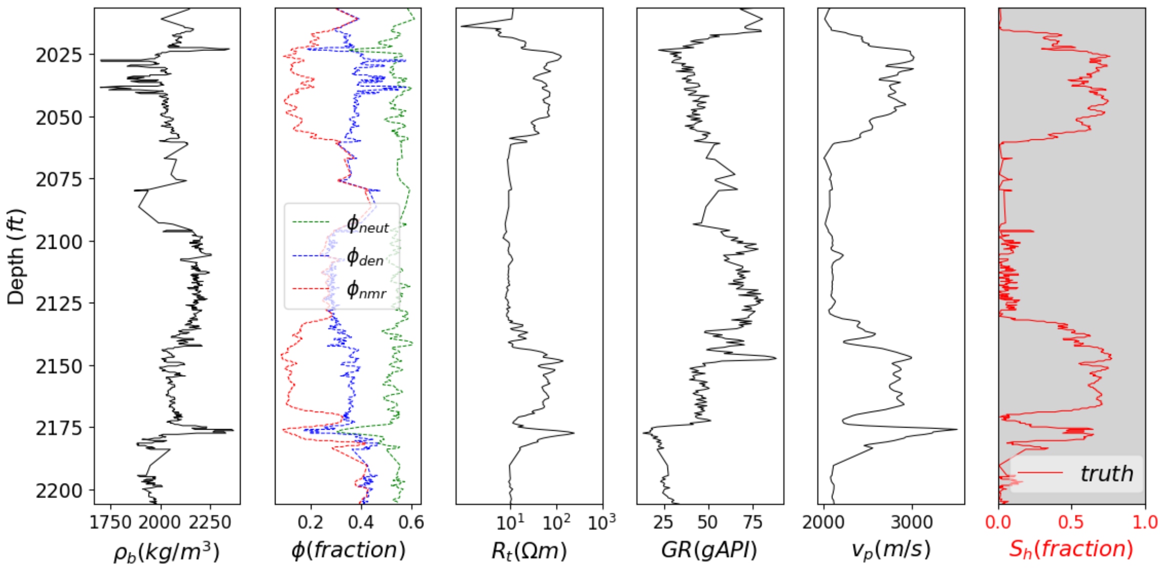

Figure 2: Relevant well-logs from Mt. Elbert Stratigraphic Test Well. The ground truth for S_h (measured using NMR-density porosity method) is highlighted to differentiate it from the other six routine well-logs.

2.5.2. Ignik Sikumi Test Well ("Well-IGS")

A joint collaboration between ConocoPhillips (Alaska), the U.S. Department of Energy, and the U.S. Geological Survey conducted a well test in 2013 that was called 'Ignik Sikumi Test Well'. The well-log data for Ignik Sikumi Test Well, located at Alaska North Slope, was acquired using wireline logs. The six routine logs (GR, ρ_b, ϕ_{neut}, ϕ_{den}, R_t, v_p), and ground truth for S_h estimated using NMR log (ϕ_{nmr}) are shown in Figure 3. The depths of investigation range from 1775.5 ft to 2595 ft with 1131 number of data points.

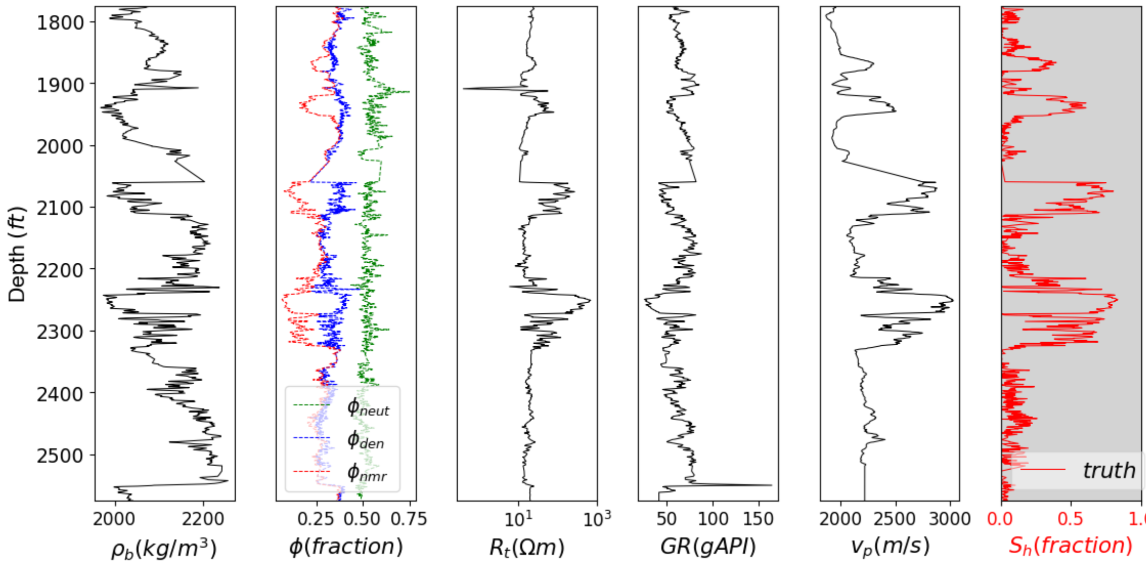

Figure 3: Relevant well-logs from Ignik Sikumi Test Well. The ground truth for S_h (measured using NMR-density porosity method) is highlighted to differentiate it from the other six routine well-logs.

The proposed method is demonstrated using the above two wells that are used interchangeably as training well and blind well in two case studies discussed later.

2.6. Accuracy of Prediction

The accuracy of the predicted well-log curve ($S_{h,predicted}$) with respect to the corresponding truth for that curve ($S_{h,truth}$) is estimated as an accuracy score (AS) using explained variance [56], which measures the accuracy of the fitted model in terms of the variation of predicted data set from its target and its special cases include coefficient of linear regression (R^2) as well as correlation coefficient:

$$Accuracy\ Score\ (AS) = 1 - \frac{var(S_{h,truth} - S_{h,predicted})}{var(S_{h,truth})} \tag{4}$$

A perfectly fitted model will have a best possible AS of 1, while as the value of AS decreases the fit of the model degrades.

While the AS can be used to assess the performance of a single ML algorithm with a given sample size, using AS to assess the performance of multiple algorithms simultaneously including their sensitivity to a size of the training data can be challenging, especially if the objective is to select a best performing algorithm while also accounting for the disparity in accuracy as a function of the training data size. In this study, this is achieved by extending the measure of AS to account for disparity in AS due to different sizes of training data through a normalized accuracy score (NAS) as follows:

$$\mathit{Normalized\ Accuracy\ Score}\ (NAS) = \underbrace{\max\{\max([AS_i]), 0\}}_{\substack{maximum\ AS \\ is\ chosen \\ only\ from \\ positive\ scores}} \times \underbrace{\{1 - var([AS_i])\}}_{\substack{normalizes\ the \\ disparity\ in \\ scores\ due \\ to\ different \\ sizes\ of \\ training\ data}} \quad (5)$$

Here, max, var represent the maximum and the variance of a sequence of accuracy scores, denoted by $[AS_i]$, where each value of this sequence (AS_i) is an accuracy score with a different sample size ($i = 1,2,3 \ldots$) of the training data, estimated for the same ML algorithm. This extended measure of accuracy score normalizes (standardizes) the disparity in accuracies due to different sizes of training dataset, hence, referred to as normalized accuracy score (NAS). A perfectly fitted model that is not affected by the size of the training data will have a best possible NAS of 1, while the disparity in performance due to size of the training data will decrease the value of NAS.

To account for the variability in performance of ML-trained algorithms due to sample sizes, 5 different sample sizes of the training datasets are considered as a percent of total data points (25%, 45%, 60%, 70%, 75%) of each well, while the remaining samples are used for validation and testing. The accuracy scores computed for these 5 different training data sizes, denoted by $[AS_i]$, are transformed into a single NAS, which is used to assess the performance of each WLC.

Cross-validation was performed on the training data to ensure the training of the model is robust and does not vary significantly if trained using different sub-parts of the training data. Cross-validation was performed using k-fold method where k-1 sets are randomly chosen as training sets while a single set is used for testing. The k-fold cross-validation allows k different combinations of training and testing data and the average of k different rankings is used to score model's training. A higher value of k indicates more number of sets for training and test, thus increased computational cost, but it does not necessarily ensure improvement in model's robustness. A value of 4 was chosen for k to maintain balance between computational cost and model robustness.

2.7. Case Studies

The proposed method is demonstrated using two case studies, which show that ML can predict S_h without calibration or limitations of the existing methods. Each case study uses a part (25%, 45%, 60%, 70%, 75% of total data size) of one well to train ML algorithms before using them to predict S_h on a 'blind well' (a second well) that confirms the robustness of the proposed method. The results from the two case studies are used to optimize the proposed method by finding the optimal WLC and optimal ML algorithm that can be used to predict S_h for any well. The remaining part of the first well (validation set and test set) in each case study can be used to double check the optimal solution for WLC, and finally a 'blind well' test will serve as a complete and independent test, depicting the performance of the proposed method.

3. Results

The method to predict S_h using ML is illustrated using two case studies, where each case study uses a part of the first well to train ML algorithms while the remaining part of the first well is used to assess the performances of WLC and different ML algorithms per standard practice. The trained algorithms are then used to predict S_h on a 'blind well' (a second well) in each case study that confirms the robustness of the proposed method.

All the 12 ML algorithms discussed earlier are trained using each of the 24 WLC shown in Table III. The results from the two case studies are used to optimize the method by finding the optimal WLC and optimal ML algorithm that can be used to predict S_h for any well.

3.1. Case Study I

For this case study, ML algorithms are trained using part of the well-logs from Well-MTE (Mt. Elbert Stratigraphic Test Well), while the blind well is chosen as Well-IGS (Ignik Sikumi Test Well). Figure 4 shows the performance of all the 24 WLC in terms of NAS for training dataset, validation dataset, and the test dataset within the same well in red, blue, and green colored legends, respectively. A NAS value less than 0 for any WLC is shown as an empty bar indicating a worse prediction of S_h by that WLC.

Figure 5 shows the performance of each WLC in predicting S_h in decreasing order along with the name of the algorithm that performs best for each WLC as the label. A corresponding result if the data contains noise is shown by Figure A-1 in Online Resource. This figure shows that the maximum accuracy achieved in predicting S_h for a blind well is ~86 % and in most cases k-Nearest Neighbors is the best performing algorithm.

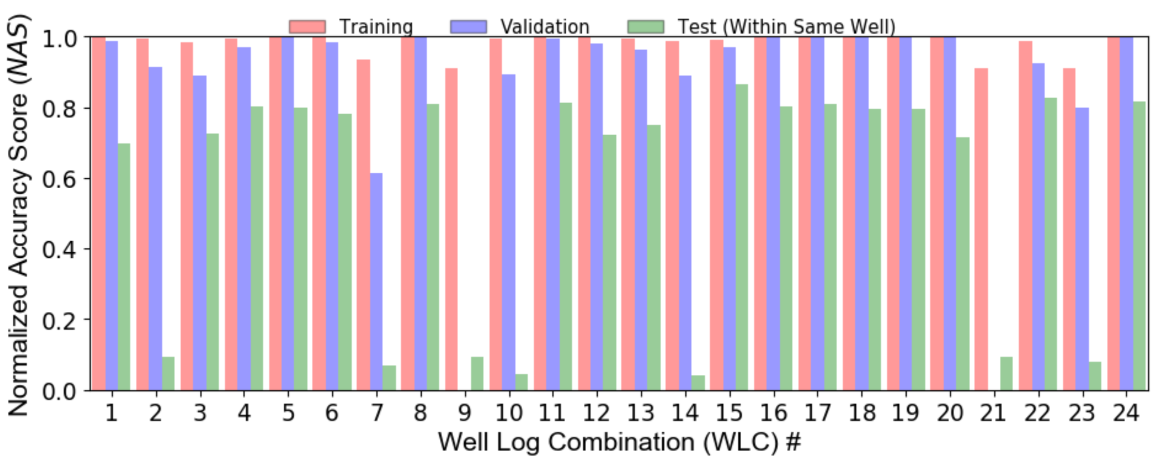

Figure 4: Accuracy of different WLC in terms of the NAS (that accounts for size of the training data) for each combination of well-logs in Well-MTE.

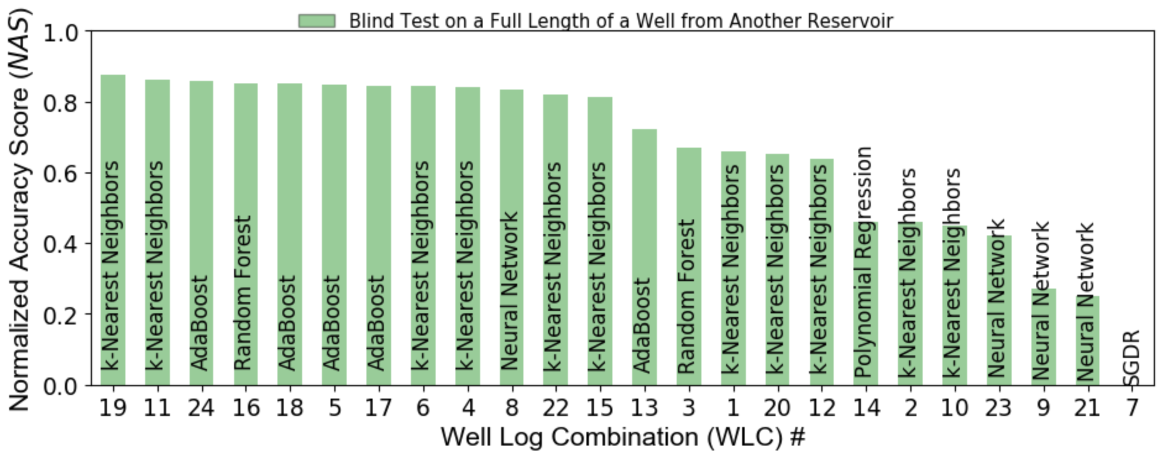

Figure 5: Accuracy of different WLC in terms of the NAS (that accounts for size of the training data) for each combination of well-logs in 'blind' well (Well-IGS). The name of the ML algorithm that performs best for each WLC is listed as a label.

3.2. Case Study II

For this case study, ML algorithms are trained using part of the well-logs from Well-IGS (Ignik Sikumi Test Well), while the blind well is chosen as Well-MTE (Mt. Elbert Stratigraphic Test Well). Figure 6 shows the performance of all the 24 WLC in terms of NAS for training dataset, validation dataset, and the test dataset within the same well in red, blue, and green colored legends, respectively. A NAS value less than 0 for any WLC is shown as an empty bar indicating a worse prediction of S_h by that WLC.

Figure 7 shows the performance of each WLC in predicting S_h in decreasing order along with the name of the algorithm that performs best for each WLC as the label. A corresponding result if the data contains noise is shown by Figure A-2 in Online Resource. This Figure shows that the maximum accuracy achieved in predicting S_h for a blind well is ~83 % and in most cases AdaBoost is the best performing algorithm.

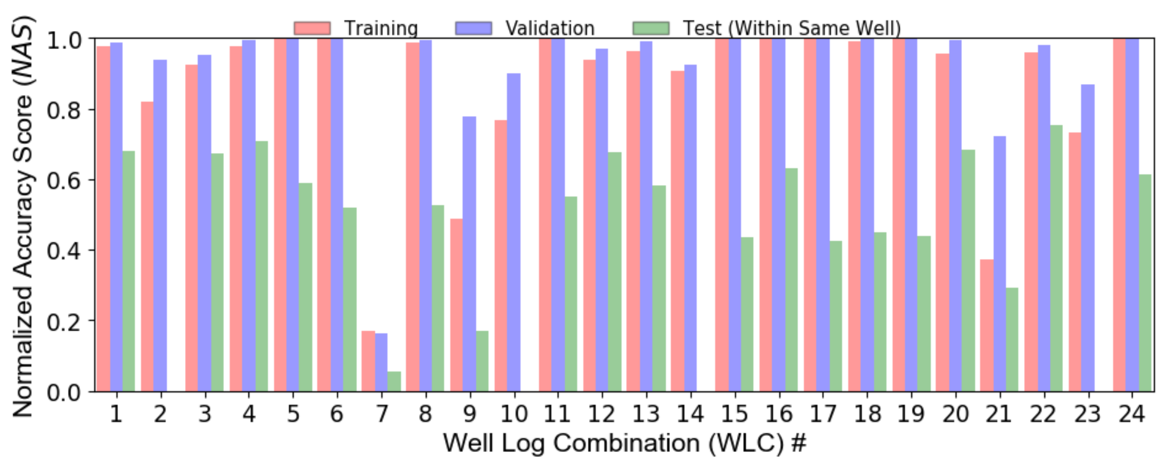

Figure 6: Accuracy of different WLC in terms of the NAS (that accounts for size of the training data) for each combination of well-logs in Well-IGS.

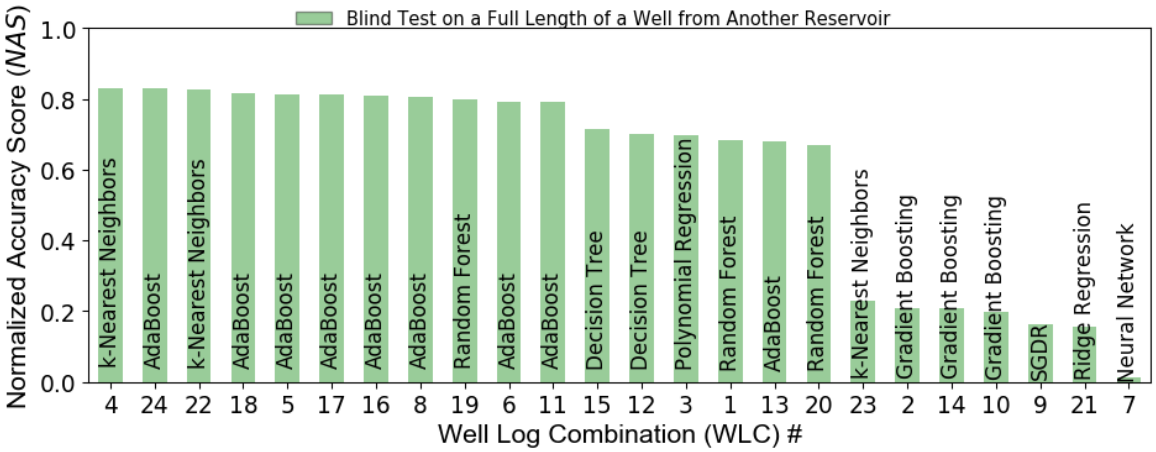

Figure 7: Accuracy of different WLC in terms of the NAS (that accounts for size of the training data) for each combination of well-logs in 'blind' well (Well-MTE). The name of the ML algorithm that performs best for each WLC is listed as a label.

3.3. Optimizing the Method

Like many other real-world measurements that are contaminated with noise, well-log data are also prone to noise that come from measurement tools (its physics and inversion methodologies) and other external sources such as borehole environment, physical variabilities and uncertainty in the surrounding rock volume, etc. Additionally, the tools used for different well-logs have different resolutions in lateral and vertical directions that result in each log having a different measure of uncertainty. Therefore, the percent of error or noise induced in well-logs is a result of all the above noted attributes and it is generally not statistically significant, for example a maximum of $\pm 1\%$ to $\pm 2\%$ for density porosity log [23], unless the borehole diameter deviates beyond a given threshold due to washouts or clay swelling that may result in relatively larger errors. The impact of noise-prone data on the performance of the proposed method is assessed by adding random white noise ($\pm 10\%$ of the noise-free data) having a uniform distribution to the well-log data, which is a common practice in the field of well-log interpretation [4] since the actual percentage of noise in each well-log is subjective as discussed.

An important point learned from the two case studies (Figure 5 and Figure 7) is that there is no distinctly unique WLC and a unique ML algorithm that performs best in both the case studies, but there are a few common WLCs that appear to be performing equally well in both the cases. Despite the lack of distinctly unique WLC and a unique ML algorithm that performs best in both the case studies, an approximately optimal WLC and an optimal algorithm are selected by finding an average of the results obtained from well-log data without noise (Table A-I in Online Resource) and the results obtained after adding random white noise ($\pm 10\%$ of the noise-free data) to the well-log data (Table A-II in Online Resource).

3.3.1. Optimal WLC

To find the optimal WLC, an average NAS (NAS_{avg}) is computed for each WLC using the values from the two case studies, which is then used to rank each WLC from 1 to 24 per the decreasing order of NAS_{avg} values as follows:

$$Rank(WLC_i) \propto \frac{NAS_I(WLC_i)+NAS_{II}(WLC_i)}{2}, \quad i = 1,2,\ldots,24 \tag{6}$$

The ranking of 24 WLCs per the NAS_{avg} are computed for results obtained with well-log data without noise and the results obtained after adding random white noise ($\pm 10\%$ of the noise-free data) to the well-log data, which are shown in section A.1.2 of the Online Resource. The final ranking of 24 WLCs as shown by Table IV, which is used to find an approximately optimal WLC, are obtained after taking corresponding average of NAS_{avg} values for each WLC from Table A-I and Table A-II in Online Resource.

The ranking for each WLC, as shown in Table IV, indicates that WLC #8 (ϕ_{neut}, ρ_b, v_p) is the optimal WLC to predict S_h with ~83% accuracy. The performance of the optimal WLC, based on the NAS_{avg}, can be verified using its performance with the validation and test datasets in first well of each case study using Figure 4 and Figure 6, respectively. Figure 4 shows that the performance of WLC #8 (ϕ_{neut}, ρ_b, v_p) is largely outperforming others in Case I, whereas Figure 6 shows that the performance of WLC #8 is almost as good as the best for validation dataset, and moderately well for test dataset. Unlike a 'blind well' that serves as a complete and independent test, the size of the data points in validation and test datasets are only part of the training well, hence the performance assessed using the validation and test datasets does not depict the holistic performance of the optimal WLC. Results for depth versus S_h as predicted using optimal WLC (ϕ_{neut}, ρ_b, v_p) for the two case studies (I and II) are presented in the Online Resource (A.1.3 and A.1.4), which show S_h predicted within the same well and along the entire length of a blind well for each case study.

Table IV: Ranking of each WLC based on its averaged performance in predicting S_h from Case Study I and Case Study II. The results shown here are obtained after taking corresponding average of NAS_{avg} values for each WLC without noise and with added noise to the well-log data (given by Table A-I and Table A-II in Online Resource).

Rank	***WLC #***	***Well-Logs***	$\boldsymbol{NAS_{avg} = \dfrac{\underbrace{NAS_{avg}}_{without\ noise} + \underbrace{NAS_{avg}}_{with\ noise}}{2}}$

1	8	$\phi_{neut},\ \rho_b,\ v_p$	0.8296
2	16	$\phi_{den},\ \rho_b,\ R_t,\ v_p$	0.8290
3	6	$\phi_{den},\ \rho_b,\ v_p$	0.8285
4	22	v_p	0.8281
5	24	$GR,\ \phi_{neut},\ \phi_{den},\ \rho_b,\ R_t,\ v_p$	0.8253
6	19	$\rho_b,\ R_t,\ v_p$	0.8248
7	4	$R_t,\ v_p$	0.8235
8	5	$\phi_{neut},\ \phi_{den},\ \rho_b,\ R_t,\ v_p$	0.8222
9	18	$\phi_{neut},\ \rho_b,\ R_t,\ v_p$	0.8221
10	11	$GR,\ \phi_{den},\ \rho_b,\ R_t,\ v_p$	0.8205
11	17	$GR,\ \phi_{neut},\ \rho_b,\ R_t,\ v_p$	0.8126
12	15	$GR,\ v_p$	0.7471
13	13	$GR,\ R_t$	0.6926
14	12	$\phi_{neut},\ \rho_b,\ R_t$	0.6825
15	1	$\phi_{den},\ R_t$	0.6784
16	3	R_t	0.6780
17	20	$\phi_{den},\ \rho_b,\ R_t$	0.6735
18	14	$GR,\ \rho_b$	0.3410
19	2	$GR,\ \phi_{neut},\ \rho_b$	0.3326
20	23	GR	0.3288
21	10	$GR,\ \phi_{neut}$	0.3249
22	21	ϕ_{den}	0.2129
23	9	ρ_b	0.1828
24	7	ϕ_{neut}	0.0095

Optimal WLC and Other Closest Options

Although WLC #8 is considered an optimal WLC based on the NAS_{avg} given in Table IV, the next ten WLC (Rank 2 through 11) following the optimal WLC perform almost similar to the optimal WLC such that the accuracy of prediction with any of these WLC is above 81%. Therefore, a different WLC can be used from Rank 2 through 11 if one or more of the well-logs in the optimal WLC (#8) are not available; out of these 10 other WLC, 4 of them contain 3 well-logs or less, which are WLC #16 ($\phi_{den},\ \rho_b,\ v_p$), #22 ($v_P$), #19 ($\rho_b,\ R_t,\ v_p$), and #4 ($R_t, v_p$), respectively.

Table IV shows a single log v_p (WLC #22) as the 4th best option to predict S_h with an accuracy of 82.81%; in other words, the P-wave velocity log alone predicts S_h better than the conventional acoustic velocity method that has been shown to predict S_h with less than 75% accuracy [22,23,36]. The reason ML-based method using only v_p log is able to predict so well in both case studies is possibly because both Well-MTE and Well-IGS belong to similar geologic basin (Eileen Gas Hydrate Trend, Alaska North Slope) that share similar lithological characteristics, facies, and possibly hydrate morphology. However, if the two wells from the two case studies were to come from two different basins, *e.g.* Alaska North Slope and the Kumano forearc basin of the Nankai Trough, a single v_p log by itself may become less effective in predicting S_h because of the different characteristics of the two basins that may be better captured by the combination of porosity (ϕ_{neut} or ϕ_{den}), bulk density, and v_p logs.

Further, sensitivity analyses of different combinations of these logs as depicted by NAS_{avg} (in Table IV, Table A-I and Table A-II) reveal that performance of analogous WLC with ϕ_{neut} and ϕ_{den} differ by less than 1% of accuracy, which indicates that either of the two porosity logs can be replaced by the other.

3.3.2. Optimal Algorithm

For the optimal WLC (#8), the ML algorithms that performed the best in Case I and Case II are Neural Network and AdaBoost, respectively, as shown by Figure 5 and Figure 7, respectively. However, on a closer inspection of the accuracy scores for all 12 ML algorithms with WLC (#8) as shown in Figure 8, there are four other algorithms (highlighted by dashed blue line) that perform almost as well as the best (Neural Network and AdaBoost, highlighted by solid blue line) in each Case I and Case II, although not all of these algorithms perform equally well in both case studies; assessing the performance of each of algorithm from Figure 8 across the two case studies reveals three ML algorithms (Neural Network, SGDR, and Ridge Regression) exhibit consistent performance in terms of their accuracy scores as well as insignificant variability in their scores with five different sizes of the training data. However, further investigation of these three algorithms based on similar results when noise is added to the well-logs (Figure A-11 in Online Resource) shows that the performance of Ridge Regression is severely affected by the size of the training data when noise is added to the well-logs (Figure A-11a in Online Resource), and Neural Network may hold a slight edge over SGDR as its performance is not affected as much as SGDR by the size of the training data (Figure A-11a in Online Resource).

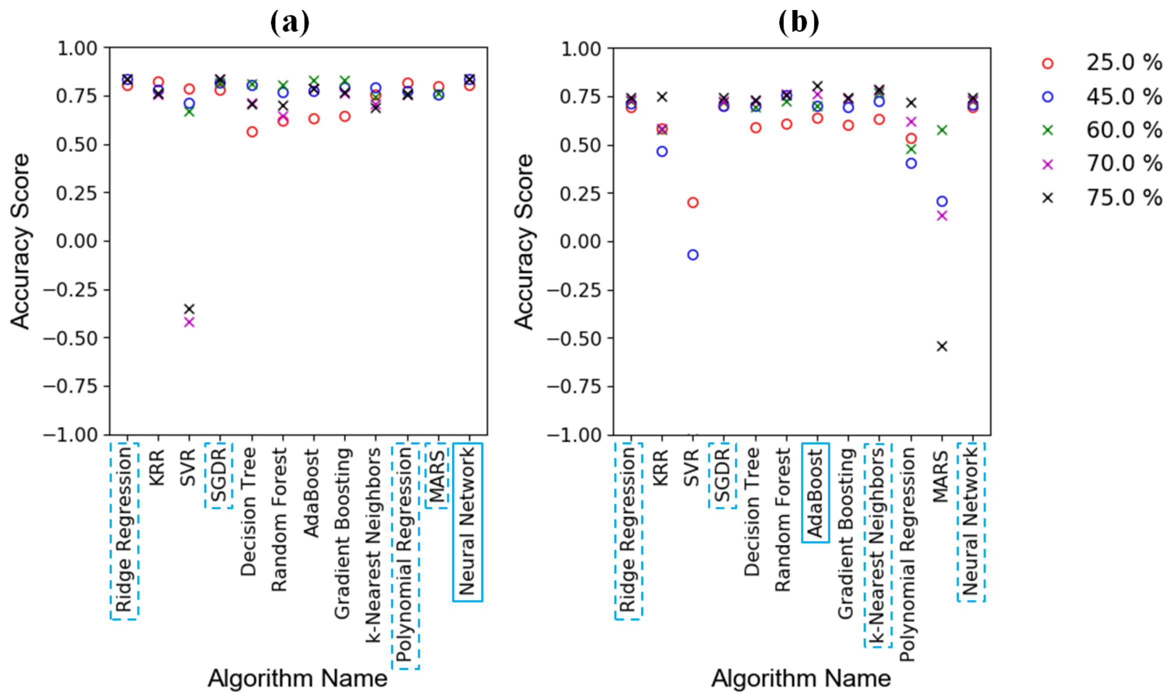

Figure 8: Accuracy scores of all the ML algorithms in predicting S_h for the optimal WLC (#8) as a function of 5 different sizes (25%, 45%, 60%, 70%, 75%) of the training data, for (a) Case Study I, (b) Case Study II. The best performing algorithm in each case study is highlighted by solid blue line, while four other algorithms that perform almost as well as the best in each case study are highlighted by dashed blue line.

A general rule of thumb for Neural Networks suggests that size of the training data should be at least 30 times the size of the weights [57], where the size of the weights in this study is denoted by the number of well-logs in any WLC. Therefore, this means the optimal WLC (#8) with 3 well-logs would require a minimum of 90 samples for each of the 3 well-logs. The minimum sample sizes of the training data for the two case studies were 206 and 283 (25% of the total data size in each case study), respectively, which exceed minimum 90 samples required by Neural Network for the optimal WLC (#8).

4. Discussions

One of the important issues with application of ML is reusing previously trained algorithm for future tasks. Vast majority of studies for application of ML in geoscience do not test the previously trained ML algorithm on 'blind' targets, meaning any target that is not associated in any form with the training of the ML algorithm; in ML parlance, this is also known as *zero-shot prior* or *zero-shot learning* [58], which means using a previously trained algorithm on an unrelated future target without retraining. The approach adopted in this study ensures that the optimal algorithm used to predict S_h does not require retraining when predicting S_h for an unrelated ('blind') well; in other words, the proposed method is a ready-to-use trained ML algorithm that can predict S_h for any unrelated target well without retraining.

It can be observed from Table IV that the performance of any particular WLC does not seem to be related to the number of logs present in that WLC. This probably means that ML is inherently performing better with those logs that have strong underlying dependency with S_h, because addition of any log that has no (or weak) relationship with S_h may attenuate the underlying dependency that any ML algorithm attempts to search; another way to interpret this is that it could be possible to further strengthen a reasonably good underlying dependency of any single well-log with S_h by the addition of other well-logs. Even though the WLC #22 with a single log (v_p) is able to predict S_h very well in both case studies with similar geologic basin (Alaska North Slope), using only v_p to predict S_h may become less effective if the two wells used for training and prediction are from different basins; this is because characteristics of two different basins may be better captured by the WLC of porosity (ϕ_{neut} or ϕ_{den}), bulk density (ρ_b), and v_p logs.

4.1. Comparing Method's Accuracy

The average S_h prediction accuracy of the two case studies is ~84% based on NMR-density porosity method discussed earlier, which is a standard practice when comparing the accuracy of existing methods [22,23,36,59]. The accuracy of existing methods (Archie's equation and acoustic velocity method) in predicting S_h was investigated by Kleinberg et al. [36] for Mallik 5L-38 gas hydrate production research well, and by Kumar et al. [22] and Collett and Lee [23] for Mt. Elbert Stratigraphic Test Well, which showed the accuracy of Archie's equation and the acoustic velocity method (using several models) lies largely between 50% to 75%. As noted in Kleinberg et al. (2005), Kumar et al. (2009) and Collett and Lee (2012) that compared the S_h predictions by the two methods, Archie's method required calibrating empirical parameters by matching the ground truth, and the acoustic velocity method required assumed values of mineralogical composition along with few consolidation parameters. The accuracy of existing methods in predicting S_h was further investigated on gas hydrate field in Krishna-Godavari Basin by Shankar et al. [59] who found that physics-based methods can either highly overestimate or highly underestimate S_h. Besides being more accurate than the existing two methods, the proposed method does not rely on well-specific calibration, which determines the accuracy of Archie's equation and the acoustic velocity method that depend on calibration of empirical parameters and accurate value of mineralogical compositions, respectively.

Archie's method is largely equivalent to curve fitting, since it requires S_h data to calibrate its parameters, and ideally the calibrated parameters are applicable only to the same well from which S_h data was used for calibration. In other words, a fair comparison of accuracy of the proposed method to Archie's method would be equivalent to comparing the ML predicted S_h for the training part of the well, which gives an almost 100% accuracy as shown by pink legends in Figure 4 and Figure 6.

Although the results show that the predictions for S_h is ~84 % in each of the two case studies despite a significant difference in the dataset sizes of two wells (822 data points in Well-MTE and 1131 data points in Well-IGS) used to train the algorithms, the accuracy value of ~84% could possibly increase or decrease when estimating S_h for a new target well and/or if the method is re-trained using some other well. This suggests that there is a scope to further increase the accuracy of prediction of S_h for any well based on the size and quality of the dataset used in training the algorithm. Further, the input data used in this study was not scaled (transformation through some mathematical formulations) to include additional features, however, it might be possible to further improve on the accuracy by scaling certain logs.

Impact of Noise

Addition of white noise ($\pm 10\%$ of the noise-free data) to the well-log data reduced the accuracy in prediction of S_h by a mere 0.05%. Although all the six well-logs used are easy to acquire, some of these logs have redundancy/overlap (e.g. ϕ_{neut}, ϕ_{den}) and/or functional dependency (e.g. ϕ_{den}, ρ_b), which helps attenuate possible presence of noise in the logs. The presence of noise-affected logs that do not provide any redundancy or functional dependency in data can affect the accuracy; this is illustrated by WLC #24 that seems to be most affected by noise (as shown by Table A-I and Table A-II), which can be due to the reason that WLC #24 contains all the six logs where some of the logs are neither redundant nor have any functional dependency.

Besides the overlap and dependency of logs that helps attenuate the noise, a large size of the well-log dataset also helps attenuate the effect of noise. For example, per the rule of thumb, a minimum number of data required for Neural Network with 3 well-logs is 90 samples per each well-log, therefore, a well-log dataset that is several times larger than 90 samples would help attenuate the effect of noise.

4.2. NEURAL NETWORK VS. OTHER LEARNING ALGORITHMS

The performance of Neural Network is often based on the activation function (σ), which helps introduce many forms of non-linearity in a Neural Network that distinguishes its capability from linear regression. Activation function can take many forms such as sigmoidal functions (e.g. arc tan), identity functions (e.g. linear), rectified linear functions, etc., and its choice is often based on trials to assess their performance. In this study, the activation function that performed the best in predicting S_h with optimal WLC is linear (identity function), while other choices of activation function, such as the rectified linear function, performed relatively worse. A Neural Network formed using an identity activation function is equivalent to a Neural Network with a single hidden layer, which means adding more than one hidden layer would not make a difference to its performance. Although SGDR and Neural Network perform largely similar to each other in terms of predicting S_h for 'blind' well without retraining, Neural Network may hold a slight edge over SGDR due to the fact that its performance is not affected as severely as SGDR with the size of the training data.

4.3. Capturing Physics: Explicit Models vs. Implicit Data-Based Learning

Mechanistic or empirical models, used to quantify petrophysical properties by processing well-log measured variables and other parameters obtained from core samples, are explicit analytical expressions that help understand the relationship between the input and output parameters. As discussed earlier, the practical utility served by these models comes with limitations that are either often overlooked or traded by their ease of use. The assumptions attached to these explicit models cannot be met by the complexity exhibited by gas hydrate reservoirs, besides the natural heterogeneity of the porous medium in terms of mineralogy, pore sizes and shapes, hydrologic properties, fluid properties, etc. [60]. The heterogeneity in these variables introduces nonlinearities that are extremely difficult to capture through an explicit, physics-based or empirical, model. For example, one of the main requirements to use Archie's law is that the reservoir must not be composed of thin laminated layers [34], but despite the common occurrence of gas hydrate reservoirs as thin layers interspersed with sand or shale sequences, such as NGHP site in India [61] and Eastern Nankai Trough of Japan [62], Archie's law has gained wide acceptance in gas hydrate reservoirs [35] possibly because of its ease of use as a single analytical equation that requires only R_t well-log. Besides, the empirical parameters (a, m, n) in Archie's law are well-specific, which means considering a universal value of these parameters with some uncertainty for all hydrate reservoirs [35] will introduce significant errors in estimated S_h.

Thus, the method proposed in this study to predict S_h using ML is an efficient and effective method that implicitly captures the porous media heterogeneities and nonlinearities depicted through well-logs while learning the underlying relationship between different variables.

The method can be further adapted to predict other important parameters like irreducible water saturation (S_{wr}), which is an important parameter in numerical assessment of gas hydrate reservoir performance [63]. Our preliminary results to predict S_{wr} using ML reveal that prediction of S_{wr} within the same well is satisfactory, but the method may need at least one modification for accurate prediction of S_{wr} in a blind well; this modification may include an additional step to perform physics-motivated feature engineering that may make it easier for the ML algorithms to capture the underlying pattern between input variables and S_{wr}.

4.4. Limitations and Ideas for Further Investigations

The method proposed in this study was trained using wells that are hosted on permafrost type hydrates hosted in lithified rocks where oilfield-type wireline logs are used. A potential idea for further investigation could involve verifying the proposed method for marine hydrates hosted in unlithified sediments where the data comes from logging while drilling measurements.

Unlike physics-based methods that have underlying assumption for the type of hydrate morphology, the proposed method using ML does not assume any type of underlying hydrate morphology; a potential idea for future investigation could verify whether the proposed method is equally applicable to gas hydrates reservoirs with different types hydrate morphologies, such as vein, nodule, or massive.

5. Summary and Conclusion

This study proposed a method that predicts gas hydrate saturation (S_h) for any well using porosity (neutron or density), bulk density, and P-wave velocity well-logs with Neural Network (or SGDR) without any well-specific calibration and/or other aforementioned shortcomings of the existing two methods (Archie's and seismic). The method is developed by examining the underlying dependency between S_h and different WLCs chosen from 6 routine logs with 12 different ML algorithms. The accuracy and robustness of the proposed method is demonstrated using two case studies (without and with added synthetic noise to the well-log data) and the results are used to obtain an approximately optimal WLC and an optimal algorithm that can be used to predict S_h for any target well. Besides the optimal WLC (ϕ_{neut}, ρ_b, v_p), there are 10 other WLC from Rank 2 through 11 that perform almost very similar to the optimal WLC and can be used if one or more of the well-logs in the optimal WLC are not available. Additionally, analyses of different WLC reveal that replacing porosity logs (ϕ_{neut} with ϕ_{den} or vice-versa) affects the accuracy by less than 1%, suggesting that either of the two porosity logs can be replaced by the other.

The method proposed in this study is free of shortcomings of the existing methods, which include i) well-specific calibration of empirical exponents in Archie's equation, ii) assumption of an ideal pore morphology for gas hydrates in the acoustic velocity method, iii) presence of unknown mineralogy and bulk modulus terms in the acoustic velocity method. The approach presented in this study to predict S_h has a potential for further improvement in accuracy, and it can be adapted with some tuning to predict other variables of interest that cannot be measured directly or are difficult to obtain, such as irreducible water saturation.

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Disclaimer

References

1. Coates, G.R.; Xiao, L.; Prammer, M.G. *NMR logging: principles and applications*; Haliburton Energy Services, 1999;
2. Mohaghegh, S.; Richardson, M.; Ameri, S. Virtual Magnetic Imaging Logs: Generation of Synthetic MRI Logs from Conventional Well Logs.; Society of Petroleum Engineers, 1998.
3. Jamshidian, M.; Hadian, M.; Zadeh, M.M.; Kazempoor, Z.; Bazargan, P.; Salehi, H. Prediction of free flowing porosity and permeability based on conventional well logging data using artificial neural networks optimized by Imperialist competitive algorithm – A case study in the South Pars gas field. *Journal of Natural Gas Science and Engineering* **2015**, *24*, 89–98, doi:10.1016/j.jngse.2015.02.026.
4. Li, H.; Misra, S.; He, J. Neural network modeling of in situ fluid-filled pore size distributions in subsurface shale reservoirs under data constraints. *Neural Comput & Applic* **2019**, doi:10.1007/s00521-019-04124-w.
5. Akkurt, R.; Conroy, T.T.; Psaila, D.; Paxton, A.; Low, J.; Spaans, P. Accelerating and Enhancing Petrophysical Analysis With Machine Learning: A Case Study of an Automated System for Well Log Outlier Detection and Reconstruction.; Society of Petrophysicists and Well-Log Analysts, 2018.
6. Wu, P.-Y.; Jain, V.; Kulkarni, M.S.; Abubakar, A. Machine learning-based method for automated well-log processing and interpretation. In *SEG Technical Program Expanded Abstracts 2018*; SEG Technical Program Expanded Abstracts; Society of Exploration Geophysicists, 2018; pp. 2041–2045.

7. Alvarado, V.; Ranson, A.; Hernandez, K.; Manrique, E.; Matheus, J.; Liscano, T.; Prosperi, N. Selection of EOR/IOR Opportunities Based on Machine Learning.; Society of Petroleum Engineers, 2002.
8. Kamari, A.; Nikookar, M.; Sahranavard, L.; Mohammadi, A.H. Efficient screening of enhanced oil recovery methods and predictive economic analysis. *Neural Comput & Applic* **2014**, *25*, 815–824, doi:10.1007/s00521-014-1553-9.
9. Tarrahi, M.; Afra, S.; Surovets, I. A Novel Automated and Probabilistic EOR Screening Method to Integrate Theoretical Screening Criteria and Real Field EOR Practices Using Machine Learning Algorithms.; Society of Petroleum Engineers, 2015.
10. Khazali, N.; Sharifi, M.; Ahmadi, M.A. Application of fuzzy decision tree in EOR screening assessment. *Journal of Petroleum Science and Engineering* **2019**, *177*, 167–180, doi:10.1016/j.petrol.2019.02.001.
11. Noshi, C.I.; Assem, A.I.; Schubert, J.J. The Role of Big Data Analytics in Exploration and Production: A Review of Benefits and Applications.; Society of Petroleum Engineers, 2018.
12. Aizenberg, I.; Sheremetov, L.; Villa-Vargas, L.; Martinez-Muñoz, J. Multilayer Neural Network with Multi-Valued Neurons in time series forecasting of oil production. *Neurocomputing* **2016**, *175*, 980–989, doi:10.1016/j.neucom.2015.06.092.
13. Cao, Q.; Banerjee, R.; Gupta, S.; Li, J.; Zhou, W.; Jeyachandra, B. Data Driven Production Forecasting Using Machine Learning.; Society of Petroleum Engineers, 2016.
14. Ma, X.; Liu, Z. Predicting the oil production using the novel multivariate nonlinear model based on Arps decline model and kernel method. *Neural Comput & Applic* **2018**, *29*, 579–591, doi:10.1007/s00521-016-2721-x.
15. Ristanto, T. Machine Learning Applied to Multiphase Production Problems. MS Thesis, Stanford University, 2018.
16. Spesivtsev, P.; Sinkov, K.; Sofronov, I.; Zimina, A.; Umnov, A.; Yarullin, R.; Vetrov, D. Predictive model for bottomhole pressure based on machine learning. *Journal of Petroleum Science and Engineering* **2018**, *166*, 825–841, doi:10.1016/j.petrol.2018.03.046.
17. Tian, C.; Horne, R. Applying Machine-Learning Techniques To Interpret Flow-Rate, Pressure, and Temperature Data From Permanent Downhole Gauges. *SPE Reservoir Evaluation & Engineering* **2019**, doi:10.2118/174034-PA.
18. Lee, M.W.; Hutchinson, D.R.; Collett, T.S.; Dillon, W.P. Seismic velocities for hydrate-bearing sediments using weighted equation. *Journal of Geophysical Research: Solid Earth* **1996**, *101*, 20347–20358, doi:10.1029/96JB01886.
19. Helgerud, M.B.; Dvorkin, J.; Nur, A.; Sakai, A.; Collett, T. Elastic-wave velocity in marine sediments with gas hydrates: Effective medium modeling. *Geophysical Research Letters* **1999**, *26*, 2021–2024, doi:10.1029/1999GL900421.

20. Chand, S.; Minshull, T.A.; Gei, D.; Carcione, J.M. Elastic velocity models for gas-hydrate-bearing sediments—a comparison. *Geophysical Journal International* **2004**, *159*, 573–590, doi:10.1111/j.1365-246X.2004.02387.x.
21. Kumar, D.; Sen, M.K.; Bangs, N.L. Gas hydrate concentration and characteristics within Hydrate Ridge inferred from multicomponent seismic reflection data. *Journal of Geophysical Research: Solid Earth* **2007**, *112*, doi:10.1029/2007JB004993.
22. Kumar, D.; Dash, R.; Dewangan, P. Methods of gas hydrate concentration estimation with field examples. *Society of Petroleum Geophysicists* **2009**, *76*.
23. Collett, T.S.; Lee, M.W. Well Log Characterization of Natural Gas-Hydrates. *Petrophysics* **2012**, *53*, 348–367.
24. Helgerud, M.B. Wave speeds in gas hydrate and sediments containing gas hydrate: A laboratory and modeling study. *Ph.D. Thesis* **2001**.
25. Dvorkin, J.; Nur, A. Time-average equation revisited. *Geophysics* **1998**, *63*, 460–464, doi:10.1190/1.1444347.
26. Dvorkin, J.; Nur, A.; Yin, H. Effective properties of cemented granular materials. *Mechanics of Materials* **1994**, *18*, 351–366, doi:10.1016/0167-6636(94)90044-2.
27. Dvorkin, J.; Nur, A. Elasticity of high-porosity sandstones: Theory for two North Sea data sets. *GEOPHYSICS* **1996**, *61*, 1363–1370, doi:10.1190/1.1444059.
28. Ecker, C.; Dvorkin, J.; Nur, A. Sediments with gas hydrates: Internal structure from seismic AVO. *GEOPHYSICS* **1998**, *63*, 1659–1669, doi:10.1190/1.1444462.
29. Singh, H.; Myshakin, E.M.; Seol, Y. A Non-Empirical Relative Permeability Model for Three Phases in Hydrate Bearing Sediments. *SPE Journal* **2018**, *SPE-193996-PA*, doi:SPE-193996-PA.
30. Singh, H.; Mahabadi, N.; Myshakin, E.M.; Seol, Y. A Mechanistic Model for Relative Permeability of Gas and Water Flow in Hydrate-Bearing Porous Media with Capillarity. *Water Resources Research* **2019**, *0*, doi:10.1029/2018WR024278.
31. Dugarov, G.A.; Duchkov, A.A.; Duchkov, A.D.; Drobchik, A.N. Laboratory validation of effective acoustic velocity models for samples bearing hydrates of different type. *Journal of Natural Gas Science and Engineering* **2019**, *63*, 38–46, doi:10.1016/j.jngse.2019.01.007.
32. Archie, G.E. The Electrical Resistivity Log as an Aid in Determining Some Reservoir Characteristics. *Transactions of the AIME* **1942**, *146*, 54–62, doi:10.2118/942054-G.
33. Lu, S.; McMechan, G.A. Estimation of gas hydrate and free gas saturation, concentration, and distribution from seismic data. *GEOPHYSICS* **2002**, *67*, 582–593, doi:10.1190/1.1468619.
34. Worthington, P.F. Recognition and evaluation of low-resistivity pay. *Petroleum Geoscience* **2000**, *6*, 77–92, doi:10.1144/petgeo.6.1.77.
35. Cook, A.E.; Waite, W.F. Archie's Saturation Exponent for Natural Gas Hydrate in Coarse-Grained Reservoirs. *Journal of Geophysical Research: Solid Earth* **2018**, *123*, 2069–2089, doi:10.1002/2017JB015138.
36. Kleinberg, R.L.; Flaum, C.; Collett, T.S. Magnetic resonance log of JAPEX/JNOC/GSC et al. Mallik 5L-38 gas hydrate production research well: gas

hydrate saturation, growth habit, and relative permeability. *BULLETIN-GEOLOGICAL SURVEY OF CANADA* **2005**, *585*, 114.
37. Lee, M.W. *Models for Gas Hydrate-Bearing Sediments Inferred from Hydraulic Permeability and Elastic Velocities: USGS Scientific Investigations Report 2008-5219*; 2008; p. 14;.
38. Collett, T.S.; Lewis, R.E.; Dallimore, S.R. Mallik 5L-38 gas hydrate production research well downhole well-log and core montages. *BULLETIN-GEOLOGICAL SURVEY OF CANADA* **2005**, *585*, 111.
39. Dallimore, S.R.; Collett, T.S.; Taylor, A.E.; Uchida, T.; Weber, M.; Chandra, A.; Mroz, T.H.; Caddel, E.M.; Inoue, T. Scientific results from the Mallik 2002 gas hydrate production research well program, Mackenzie Delta, northwest territories, Canada: Preface. *Geological Survey of Canada* **2005**, *585*, 1–16.
40. Jain, V.; Saumya, S.; Vij, J.; Singh, J.; Singh, B.; Pattnaik, S.; Oli, A.; Kumar, P.; Collett, T.S. New technique for accurate porosity estimation from logging-while-drilling nuclear magnetic resonance data, NGHP-02 expedition, offshore, India. *Marine and Petroleum Geology* **2018**, doi:10.1016/j.marpetgeo.2018.11.001.
41. Hastie, T.; Tibshirani, R.; Friedman, J. *The elements of statistical learning*; Springer series in statistics New York, 2001; Vol. 1;.
42. Shawe-Taylor, D. of C.S.R.H.J.; Shawe-Taylor, J.; Cristianini, N. *Kernel Methods for Pattern Analysis*; Cambridge University Press, 2004; ISBN 978-0-521-81397-6.
43. Smola, A.J.; Schölkopf, B. A tutorial on support vector regression. *Statistics and Computing* **2004**, *14*, 199–222, doi:10.1023/B:STCO.0000035301.49549.88.
44. Robbins, H.; Monro, S. A Stochastic Approximation Method. *The Annals of Mathematical Statistics* **1951**, *22*, 400–407.
45. Breiman, L.; Friedman, J.; Stone, C.J.; Olshen, R.A. *Classification and Regression Trees*; Taylor & Francis, 1984; ISBN 978-0-412-04841-8.
46. Breiman, L. Random Forests. *Machine Learning* **2001**, *45*, 5–32, doi:10.1023/A:1010933404324.
47. Schapire, R.E.; Freund, Y. *Boosting: Foundations and Algorithms*; MIT Press, 2012; ISBN 978-0-262-01718-3.
48. Friedman, J.H. Greedy Function Approximation: A Gradient Boosting Machine. *The Annals of Statistics* **2001**, *29*, 1189–1232.
49. Dudani, S.A. The Distance-Weighted k-Nearest-Neighbor Rule. *IEEE Transactions on Systems, Man, and Cybernetics* **1976**, *SMC-6*, 325–327, doi:10.1109/TSMC.1976.5408784.
50. Pal, S.K.; Mitra, S. Multilayer perceptron, fuzzy sets, and classification. *IEEE Transactions on Neural Networks* **1992**, *3*, 683–697, doi:10.1109/72.159058.
51. Friedman, J.H. Multivariate Adaptive Regression Splines. *Ann. Statist.* **1991**, *19*, 1–67, doi:10.1214/aos/1176347963.
52. Hastie, T.J.; Tibshirani, R.J. *Generalized Additive Models*; CRC Press, 1990; ISBN 978-0-412-34390-2.

53. Bhowmick, D.; Shankar, U.; Maiti, S. Revisiting Supervised Learning in the Context of Predicting Gas Hydrate Saturation.; European Association of Geoscientists & Engineers, 2016; Vol. 2016, pp. 1–5.
54. Boswell, R.; Hunter, R.; Collett, T.S.; Digert, S.; Hancock, S.H.; Weeks, M.; Team, M.E.S. Investigation of Gas Hydrate-Bearing Sandstone Reservoirs at the "Mount Elbert" Stratigraphic Test Well, Milne Point, Alaska. **2008**, doi:10.14288/1.0041029.
55. Inks, T.L.; Agena, W.F. Successful gas hydrate prospecting using 3D seismic - A case study for the Mt. Elbert prospect, Milne Point, North Slope Alaska. **2008**, *27*, 473–477, doi:10.1190/1.3054848.
56. Good, R.; Fletcher, H.J. Reporting explained variance. *Journal of Research in Science Teaching* **1981**, *18*, 1–7, doi:10.1002/tea.3660180102.
57. Abu-Mostafa, Y.S.; Magdon-Ismail, M.; Lin, H.-T. *Learning from Data: A Short Course*; AMLBook.com, 2012; ISBN 978-1-60049-006-4.
58. Gavves, E.; Mensink, T.; Tommasi, T.; Snoek, C.G.M.; Tuytelaars, T. Active Transfer Learning with Zero-Shot Priors: Reusing Past Datasets for Future Tasks. *arXiv:1510.01544 [cs]* **2015**.
59. Shankar, U.; Gupta, D.K.; Bhowmick, D.; Sain, K. Gas hydrate and free gas saturations using rock physics modelling at site NGHP-01-05 and 07 in the Krishna–Godavari Basin, eastern Indian margin. *Journal of Petroleum Science and Engineering* **2013**, *106*, 62–70, doi:10.1016/j.petrol.2013.04.004.
60. Myshakin, E.M.; Ajayi, T.; Anderson, B.J.; Seol, Y.; Boswell, R. Numerical simulations of depressurization-induced gas production from gas hydrates using 3-D heterogeneous models of L-Pad, Prudhoe Bay Unit, North Slope Alaska. *Journal of Natural Gas Science and Engineering* **2016**, *35*, 1336–1352, doi:10.1016/j.jngse.2016.09.070.
61. Myshakin, E.M.; Seol, Y.; Lin, J.-S.; Uchida, S.; Collett, T.S.; Boswell, R. Numerical simulations of depressurization-induced gas production from an interbedded turbidite gas hydrate-bearing sedimentary section in the offshore India: Site NGHP-02-16 (Area-B). *Marine and Petroleum Geology* **2018**, doi:10.1016/j.marpetgeo.2018.10.047.
62. Yuan, Y.; Xu, T.; Xia, Y.; Xin, X. Evaluation of Gas Productivity from Layered Heterogeneity Methane Hydrate Reservoirs by Depressurisation.; Offshore Technology Conference, 2018.
63. Myshakin, E.M.; Anderson, B.J.; Rose, K.; Boswell, R. Simulations of Variable Bottomhole Pressure Regimes to Improve Production from the Double-Unit Mount Elbert, Milne Point Unit, North Slope Alaska Hydrate Deposit. *Energy Fuels* **2011**, *25*, 1077–1091, doi:10.1021/ef101407b.

Declarations

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CONFLICTS OF INTEREST/COMPETING INTERESTS

The authors declare there is no competing interest.

AVAILABILITY OF DATA AND MATERIAL

Data used in this study will be uploaded to a repository on https://edx.netl.doe.gov, pending acceptance.

CODE AVAILABILITY

The code for this study was written in Python using scikit-learn and Tensorflow as the machine learning packages.

AUTHORS' CONTRIBUTIONS

HS conceived the research idea, designed the workflow, and wrote the code. H.S., E.M.M., and Y.S. analyzed the results. H.S. wrote the manuscript, E.M.M. and Y.S. edited it. All authors have reviewed and approved the final version of the manuscript.